



OPEN Ecosystem engineering and food web stability

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Ecosystem engineering, which involves organism-triggered physical modification of the environment, is a widespread phenomenon. Despite this, the role of engineering in ecological communities remains poorly understood. This study employs a food web model to uncover the key roles of ecosystem engineering in maintaining food webs. While engineers facilitating population growth and suppressing consumers' foraging activity can help maintain complex communities with diverse species, engineering effects that suppress population growth and facilitate consumers' foraging activity can largely destabilize community dynamics. Furthermore, in the middle levels of engineering-related species within a community, **an increase in species richness can increase community stability, contrary to classical ecological prediction**. The study findings suggest that ecosystem engineering can explain biodiversity persistence in nature, but it depends on the proportion of engineering-related species and how engineering affects organisms.

Many organisms participate in the creation, modification, and maintenance of abiotic environments^{1–8}. Such organisms are called “ecosystem engineers” and the process is called “ecosystem engineering”^{9,10}. A textbook example is the beaver, which cuts and uses trees to construct dams, altering hydrology and creating wetlands that largely influence plant and animal abundance, community composition, and diversity¹¹. Trees, through their branches, backs, roots, and living and dead leaf surfaces, provide shelter, resting locations, and living space to diverse organisms^{12,13}. Phytoplankton can absorb light in the upper water column, affecting the light distribution across water column^{14,15}. Although various ecosystem engineers and forms of engineering are widely recognized today, the foundational concept traces back to Darwin's example of earthworm-induced soil formation, illustrating its historical roots¹⁶. However, despite the inherent importance of ecosystem engineering for the dynamics and structure of ecological communities, insights into the roles of engineering in the dynamics and stability of ecological communities are limited^{3,17}.

Sustaining diversity or ensuring the persistence of various species over extended periods remains a fundamental question in ecology¹⁸. Classical ecological theory indicates that complex communities with diverse species are inherently unstable¹⁹, and suggests that because it assumed random interaction network structures, there are non-random structures in natural ecosystems, which in themselves could account for the stability of complex communities^{20–26}. Nevertheless, uncovering the contradiction between ecological theory and observations is still a significant challenge in ecology^{27,28}. Ecological theoretical studies have tackled this so-called complexity–stability problem by traditionally examining the dynamics of ecological communities driven by species interactions such as predation, competition, and mutualism^{29–33}. However, in natural ecosystems, the physical structures available to organisms, as well as those created by them (potentially causing changes in the physical state of biotic or abiotic substances), enable ecosystem engineers to directly or indirectly regulate resource availability to other organisms^{9,10}. This suggests that ecosystem engineering can affect the dynamics of ecological communities by providing refugia from or increasing exposure to abiotic forcing and predation and changing the growth rates of engineered species (receiver species affected by engineering species) and the strength of species interactions¹⁰. Relatively few models have explored how engineering affects ecological communities^{3,17}. These models revealed that engineering can largely influence population persistence, population dynamics, invasion success, and environmental heterogeneity^{4,34–38}. However, most of them considered small systems comprising a few species, leaving an open question of how it works in more complex communities. A recent exceptional theoretical study explored the impact of engineering on the dynamics of ecosystem assembly based on the colonization and extinction of species within a community and showed that engineering can enhance community diversity³⁹. However, this did not trace population dynamics and not examine the effects of species diversity or community complexity on the stability or persistence of ecological communities, leaving an unanswered question of how engineering drives the population dynamics of diverse-interacting species and plays a role in community maintenance and complexity–stability relationships.

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This study presents a mathematical food web model with engineering species to explore how engineering plays a role in community stability. Consider a food web comprising N species, where a proportion C of all possible prey–predator species pairs interacts. In this system, let us introduce engineering species (the proportion p_E of randomly chosen species is defined as engineers). Next, consider the receiver species affected by the engineers (the proportion p_R of randomly chosen species is defined as receivers). The engineering effects are assumed to affect the parameters of receivers as a saturating function of engineer abundance. The target parameters are the growth rate (birth or death rate) and foraging rate. Engineers might increase and/or decrease foraging rates due to changes in abiotic environments (e.g., thermal buffering or exposure) and/or nontrophic resources (engineered biotic or abiotic structure per se; e.g., prey refuges or predator ambush locations). Changes in abiotic environments, such as thermal buffering or exposure, also affect survival rates. In addition, engineering can change survival and/or reproduction rates via changes in available abiotic resources. The directions of engineering effects on growth rates (foraging rates) are controlled by q_r (q_a), which is defined as the proportion of engineering effects decreasing growth rates (foraging rates) (i.e., $1 - q_r$ is the proportion of engineering effects increasing growth rates ($1 - q_a$ in foraging rates)). By systematically controlling the parameters p_E , p_R , q_r , and q_a , this study reveals the role of ecosystem engineering in community stability, defined as the probability that all species persist for a given time (Methods).

Results

Consider a food web without ecosystem engineers ($p_E = 0$). The classical food web model shows a middle level of stability ($p_E = 0$ in Fig. 1a). However, once some engineering species is added into the system ($p_E > 0$), it can either lose or raise its stability, depending on the proportion of receiver species p_R . When the receiver is rare, adding a few engineers largely decreases stability (e.g., $p_R = 0.1$ and 0.3 in Fig. 1a), but as engineers become dominant within the system (large values of p_E), stability increases again. Conversely, when the receiver is common, engineers have totally different effects. Adding some engineers largely increases stability, but the system dominated by the engineers loses stability again (e.g., $p_R = 0.7$ and 0.9 in Fig. 1a). The contrasting results suggest that a particular proportion of species related to engineering, including engineers and receivers within the community, can stabilize the system. One measure that represents the dominance of species related to engineering within the system is $p_E p_R$ (hereinafter called “engineering dominance”). In a middle level of engineering dominance, the system has a peak of stability ($p_E p_R$ is about 0.1 – 0.15). The stabilization effect due to engineering also depends on the engineering effect type. With higher proportions of the engineering effects on growth increment and foraging reduction, engineering has a stabilizing effect (Fig. 1b). The engineering effects on stability do not depend on the network types of the food web, random, or cascade model (Fig. S1).

Engineering can markedly affect the complexity–stability relationship. Without engineering ($p_E = 0$), increasing a community complexity, species richness ($p_E = 0$ in Fig. 2), tends to decrease stability. However, engineering with a higher proportion of growth increment and foraging reduction can completely change the complexity effect on stability. Increasing species richness can elevate stability in communities with a particular combination of p_E and p_R (Figs. 2a–c, S2, and S3). Figure 3 shows that such positive complexity–stability relationship appears in a moderate range of engineering dominance. In the cascade model, stabilization by increasing species richness is also observed (Figs. S4, S5). Overall, these findings indicate that engineering through its effect on growth increment and foraging reduction can increase stability for mid-level engineering dominance and can create a positive species diversity effect on community stability.

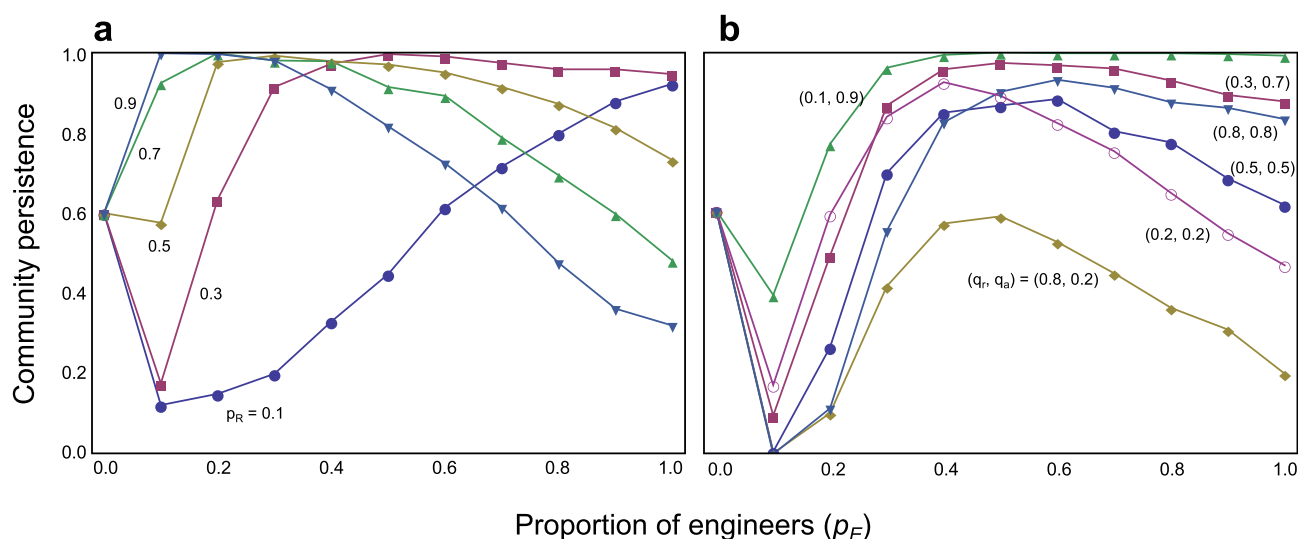


Fig. 1. Effects of engineering on community stability. Change in engineering effect on stability with varying (a) proportion of receiver species p_R and (b) proportion of different types of engineering effects (decreasing growth rates q_r and foraging rates q_a). In (a), $q_r = 0.2$ and $q_a = 0.8$. In (b), $p_R = 0.3$. A random food web is assumed. $N = 60$ and $C = 0.1$.

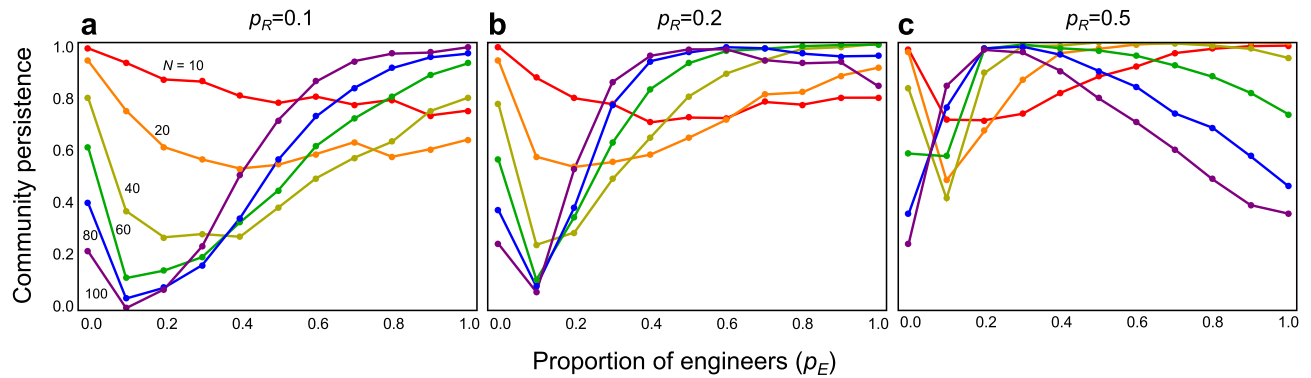


Fig. 2. Effects of engineering on a complexity–stability relationship. Colors represent the levels of species richness. **(a)** $p_R=0.1$. **(b)** $p_R=0.2$. **(c)** $p_R=0.5$. A random food web is assumed. The parameters are $q_r=0.2$, $q_a=0.8$, and $C=0.1$.

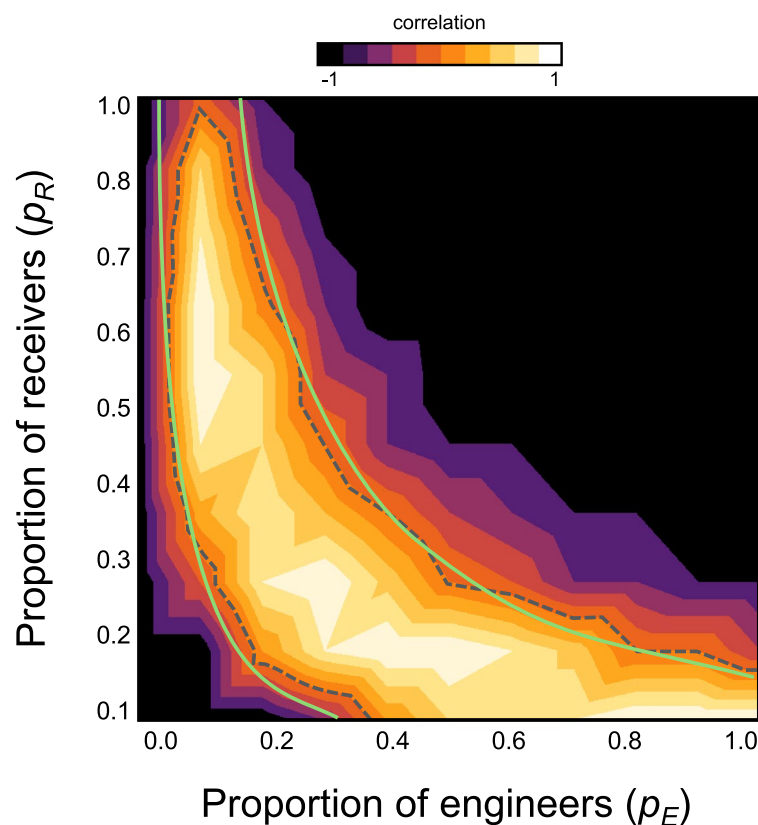


Fig. 3. Effects of engineering dominance on a complexity–stability relationship. Kendall's Tau was calculated between ($N=20, 40, 60, 80, 100$) and each persistence value, depending on combinations of parameters (p_E and p_R), serving as an index of the relationship between complexity and stability. Data for $N=10$, exhibiting higher stability, were excluded due to the potential masking effect of a “concave upward pattern” on the positive correlation, despite a strong positive complexity–stability relationship. Contours depict Kendall's Tau values ranging from -1 to 1 , where negative values indicate a negative complexity–stability relationship, positive values indicate a positive complexity–stability relationship, and a dashed line represents no correlation (0). The left and right green lines denote different engineering dominance values ($p_E p_R$), 0.03 and 0.16 , respectively.

Discussion

The model predicted that (i) ecosystem engineering with growth increment and foraging reduction is key for stabilizing food webs, (ii) moderate engineering dominance enables the system to stabilize, and (iii) species diversity can contribute to community stability under moderate engineering dominance. The results showed that engineering effects on community stability depend on the engineering effect type, engineering-related species number, and community complexity. This suggests that ecosystem engineering is a double-edged sword

in terms of community stability: low to moderate engineering levels have a positive influence on stability, while high engineering levels lead to a negative correlation.

Intermediate-level engineering dominance within an ecological community may contribute to community maintenance. This prediction is similar to the intermediate disturbance hypothesis⁴⁰ that predicts that the highest species diversity will occur at intermediate-level disturbance (If the disturbance is too mild or too rare, then patches will approach equilibrium and be dominated by a few species that can outcompete all others. If the disturbance is too harsh or too common, only a few species resistant to the disruption will persist). The engineer and its influence on habitat and thereby community structure can be treated as a disturbance, and intermediate levels of engineering are considered to represent an intermediate disturbance regime. In low engineering dominance, only some particular species can be more (or less) abundant owing to engineering effects. However, this can increase prey vulnerability or have a negative effect on the competitor sharing resources. This engineering effect was also suggested by a previous study using a colonization–extinction assembly model³⁹. In the assembly model³⁹, community network nodes include non-engineering entities, engineering entities, and abiotic entities made by engineers (modifiers affecting species). Species colonize the community stochastically if all necessary entities are present, including at least one entity they consume. A species faces extinction risk if it's not the strongest competitor for at least one resource. Engineer-made modifiers may persist after the engineer goes extinct, aiding species temporarily. Rare engineers/modifiers offer extra resources for consumers, heightening prey vulnerability and leading to more competitive exclusion of prey and primary extinctions (conversely, rich engineers/modifiers expand niche space, reducing competition and suppressing primary and secondary extinctions). On the other hand, how does higher engineering dominance affect to the stability of population dynamics? In higher engineering dominance, many species can experience an increase in population size due to positive feedback, where they mutually promote each other's growth. Consequently, interspecies interactions are intensified, leading to a significant destabilizing effect even when accounting for another perspective of engineering (i.e., reduction of foraging rates). This destabilization effect is not shown in the assembly model without population dynamics³⁹.

The present study suggests that ecosystem engineering can play a key role in maintaining larger systems. More specifically, at moderate-level engineering dominance, complex systems with numerous species can be more stable than simple systems with few species. Recent theoretical studies have also examined the effects of engineering on the stability of larger communities. One theoretical study⁴¹ showed that habitat modification, by altering the ease of establishment for another species, can create habitat heterogeneity, allowing the coexistence of multiple competing species. Another simulation study⁴² demonstrated that chemically mediated strong interactions among bacteria, caused by themselves, prevent the coexistence of diverse species, and weak chemical interactions are necessary for maintaining species diversity. These findings suggest that heterogeneity (created by diverse species with different characteristics on modifying the abiotic environments) or weak interactions (weak modification of abiotic environments) play a key role in stability of communities with engineers. These two factors are major role for community stability irrespective of the presence of engineers^{43–45}. In the present model, the both factors play crucial roles in stability. It is evident that foraging reduction leads to weak interactions. The stabilization effect of foraging reduction by engineering is also supported by simple food web theory¹⁷. The previous study considered a three trophic food chain. Analogous to the present study, the earlier model incorporates population dynamics, with engineer effects capable of modifying resource growth rates (carrying capacity) and consumption rates. Within this framework, researchers examined the impact of engineer species on the stability of population dynamics by exploring various combinations of which species, at each trophic level, function as engineer species and which species are affected by engineer. Overall, this study illustrates that reductions in foraging activity and resource growth, induced by engineers, tend to promote system stability. On the other hand, intermediate engineering dominance would create heterogeneity (parameter variation among species). Lower engineering dominance decreases the evenness of parameter values among species, while higher engineering dominance decreases parameter variation among species. Thus, communities with intermediate engineering dominance have heterogeneous and weak interactions, which can prevent the dominance of certain species and allow the coexistence of multiple species. The present model prediction supports the previous argument¹⁰ and natural observations^{1,46,47}. In both animal and plant ecosystems, animal or plant engineers in each system positively affect species diversity^{1,46,47}. Furthermore, engineering effects are stronger in the tropics than at higher latitudes¹. This is probably because in tropical regions, where there are numerous species and interactions, the creation or modification of habitats by engineers may aid in reducing competition and predation pressures on resident species¹. An additional test of the theory is to determine the relationship between engineering dominance and community stability.

The study findings have important implications for conservation. Humans are also ecosystem engineers. What does anthropogenic ecosystem engineering affect ecosystems? Empirical data suggest a negative effect of anthropogenic engineering on species diversity¹. For example, forest management and conventional farming are reported to largely decrease species richness⁴⁸. Forest management such as clearcut with tree species can decrease microhabitat availability for more or less several substrate-dependent taxa including saproxylic beetles, carbid beetles and fungi (because the detailed features of dead wood, the presence of large logs and snags and the presence of different decay stages are the key elements as resources and habitats for these taxa)⁴⁹ and would decrease their growth rates. Conventional farming can decrease the growth rates of weed by herbicides, resulting in subsequent reduction of insects and birds depending on such plant species. Pesticides can not only reduce populations of targeted pest insects but also impact the populations of their predators. If the anthropogenic engineering may generally reduce the growth rates of organisms, the communities are predicted to be unstable. A key feature of anthropogenic engineering is its large-scale impact and influence on diverse species. Thus, it implies that many species of engineering receivers p_R can bias engineering dominance toward higher values. The bias of engineering dominance may critically destabilize a complex ecosystem with diverse species by shifting the

positive complexity–stability relationship toward the negative one. Furthermore, the shift of complexity–stability relationship implies that a simplified system with species loss is more stable, making recovery to larger ecosystems difficult.

Materials and Methods

Consider a random community in which a pair of species i and j ($i, j = 1, \dots, N$) is connected by any interaction with a probability of C , which is defined as the proportion of realized interaction links L of the possible maximum interaction links, $L_{\max}$, of a given network model ($L = CL_{\max}$). To examine the generalization of the main results, cascade food webs were also tested. In the cascade model, for each pair of species, $i, j = 1, \dots, N$, where $i < j$, species i never consumes species j , whereas species j may consume species i . The maximum link number $L_{\max}$ is calculated using $N(N-1)/2$ in both the random and cascade models. The community model is defined by the following ordinary differential equation:

$$dX_i/dt = (r_i - s_i X_i + \sum_j M_{ij} X_j) X_i \quad (1)$$

where X_i is the abundance of species i , r_i is the intrinsic rate of change in species i , s_i is the density-dependent self-regulation of species i , and M_{ij} is the interaction coefficient between species i and j ($j \neq i$). The growth rates are defined as $r_i = r_{0i} B_i$, where r_{0i} is the intrinsic growth rate of species i in the absence of engineering effects; and B_i is the engineering effect on the growth rate of species i (see below). Interaction coefficients are defined as $M_{ij} = e_{ij} A_{ij}$ and $M_{ji} = -A_{ij}$, where $A_{ij} = \Gamma_i a_{ij} / (1 + \sum_{k \in \text{resources of species } i} h_{ik} a_{ik} X_k)$ (a type II functional response is assumed); a_{ij} is the consumption rate of resource species j by species i ; h_{ij} is the handling time; e_{ij} (< 1) is the conversion efficiency; and Γ_i is the engineering effect on the consumption rate of species i (see below).

This study incorporated the engineering effects on parameters, intrinsic growth rates, and consumption rates into the model as a saturating function of engineer abundance: $B_i = \sum_j (\beta_{Eji} X_{Ej} + E_{j0}) / (X_{Ej} + E_{j0})$ and $\Gamma_i = \sum_j (\gamma_{Eji} X_{Ej} + E_{j0}) / (X_{Ej} + E_{j0})$, where X_{Ej} is the abundance of ecosystem engineer species j ; E_{j0} is the half saturating abundance of the engineering effect; and β_{Eji} and γ_{Eji} represent the direction and strength of the per capita engineering effect of engineer species j on the growth and consumption rates of species i , respectively¹⁷. If β_{Eji} (γ_{Eji}) < 1 , engineer species j decreases the growth rate (consumption rates) of species i , while if β_{Eji} (γ_{Eji}) > 1 , engineer species j increases the growth rate (consumption rates) of species i . β_{Eji} (γ_{Eji}) $\neq 1$ is assumed in the developed model. The engineering effect of each engineer species j saturates at the values β_{Eji} and γ_{Eji} when the engineer species is abundant. In this model, the engineering effects of multiple engineering species on any species i are additive. For example, if the engineering effects of all engineer species on the growth rate of a focal species i are positive ($\beta_{Eji} > 1$), the growth rate of species i increases with an increasing number of engineer species. However, if not all engineer species have negative effects on species i ($\beta_{Eji} < 1$), the increases in the engineer species can both positively and negatively affect the population growth of species i . This is also true for consumption rates. Engineering effects may affect the intrinsic growth rate by increasing or decreasing the survival rate of a receiver or increasing (decreasing) reproduction or growth rate by newly using other external resources (losing any external resource).

Consider the incorporation of engineer species into a food web. The proportion of engineer species within the food web is defined as p_E (i.e., $1 - p_E$ is the proportion of non-engineer species). The engineer species, $p_E N$, affects the receiver species, $p_R N$, where p_R is the proportion of species that receive the engineering effects (i.e., $1 - p_R$ is the proportion of nonreceiver species). Here, the receiver species is also assumed to be an engineer species (an engineer species may be affected by other engineer species and/or by itself). The engineer and receiver species are randomly determined. Engineering effects may increase or decrease growth and consumption rates. The directions of engineering effects on growth rates (consumption rates) are controlled by q_r (q_a), which is defined as the proportion of engineering effects decreasing growth rates (consumption rates) (i.e., $1 - q_r$ ($1 - q_a$) is the proportion of engineering effects increasing growth rates (consumption rates)). The direction in which an engineer species changes the parameters of the receivers is randomly and independently determined.

Simulation runs (using Mathematica) were iterated with randomly chosen parameter sets from a uniform distribution ($0.2 < r_{0i} < 1.0$, $0.1 < e_{ij} < 0.25$, $0.0 < a_{ij} < 0.01$, $1.0 < h_{ij} < 10$, $0.0 < E_{j0} < 0.1$, $s_i = 1.0$). When β_{Eji} (γ_{Eji}) < 1 , $0.0 < \beta_{Eji}$ and $\gamma_{Eji} < 1.0$. When β_{Eji} (γ_{Eji}) > 1 , $1.0 < \beta_{Eji}$ and $\gamma_{Eji} < 2.0$. The values of a_{ij} are within a range of empirically estimated values based on predator–prey body size ratio, B_r ⁴⁰. The consumption rate a_{ij} corresponds to the product of metabolism rate and maximum consumption rate divided by number of available resources⁵⁰. In larger communities (e.g. in communities with $N = 100$ and $C = 0.1$, average number of resources per species is around 5), the consumption rates are about within a range between 0.003 and 1.24 (mean: 0.157) in ectotherm vertebrates ($10^{-1} < B_r < 10^9$) and between 0.0009 and 1.58 (mean: 0.259) in invertebrates ($10^{-2} < B_r < 10^{11}$). Considering that the values are skewed toward small values⁵⁰, the values of a_{ij} used in this study are not far from the estimated values. Community persistence, which is used as a representative of stability, was then calculated by measuring the fraction of simulation runs in which the entire community persisted ($X_i > 0.2$ for all i) after a sufficiently long time ($t = 300$, which corresponded to the time taken for community persistence to reach an asymptote) of 500 runs.

Data availability

All data generated or analysed during this study are included in this published article.

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A.M. performed all research and wrote the entire manuscript.

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Competing interests

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Additional information

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